



中国科学技术大学

数理方程 UNIVERSITY OF SCIENCE AND TECHNOLOGY OF CHINA

Hefei, Anhui. 230026 The People's Republic of China

第八周作业

1. 解: (此处“=”指在分布意义下相等, 即对任意测试函数 $\phi(x)$ 的作用相同)

$$(1) \int_{-\infty}^{+\infty} x \delta(x) \phi(x) dx = [x \phi(x)]|_{x=0} = 0 = \int_{-\infty}^{+\infty} 0 \cdot \phi(x) dx \Rightarrow x \delta(x) = 0$$

$$(2) \begin{cases} \int_{-\infty}^{+\infty} f(x) \delta(x-a) \phi(x) dx = [f(x) \phi(x)]|_{x=a} = f(a) \phi(a) \\ \int_{-\infty}^{+\infty} f(x) \delta(x-a) \phi(x) dx = f(a) \cdot [\phi(x)]|_{x=a} = f(a) \phi(a) \end{cases} \Rightarrow f(x) \delta(x-a) = f(a) \delta(x-a)$$

$$(3) a > 0 \text{ 时, } \begin{cases} \int_{-\infty}^{+\infty} \delta(ax) \phi(x) dx \xrightarrow[t=a \cdot x]{t=x} \int_{-\infty}^{+\infty} \delta(t) \phi(\frac{t}{a}) \frac{dt}{a} = \frac{1}{a} \phi(0) \\ \int_{-\infty}^{+\infty} \frac{\delta(x)}{a} \phi(x) dx = \frac{1}{a} \phi(0) \end{cases} \Rightarrow \delta(ax) = \frac{\delta(x)}{a}$$

$$a < 0 \text{ 时, } \begin{cases} \int_{-\infty}^{+\infty} \delta(ax) \phi(x) dx \xrightarrow[t=a \cdot x]{t=x} \int_{+\infty}^{-\infty} \delta(t) \phi(\frac{t}{a}) \frac{dt}{a} = -\frac{1}{a} \phi(0) \\ \int_{-\infty}^{+\infty} \frac{\delta(x)}{-a} \phi(x) dx = -\frac{1}{a} \phi(0) \end{cases} \Rightarrow \delta(ax) = \frac{\delta(x)}{-a}$$

综上 $\delta(ax) = \frac{\delta(x)}{|a|} \quad (a \neq 0)$

$$(4) \begin{cases} \int_{-\infty}^{+\infty} \delta'(-x) \phi(x) dx = \int_{+\infty}^{-\infty} \delta'(t) \phi(-t) (-dt) = \int_{-\infty}^{+\infty} \delta'(t) \phi(-t) dt = -\frac{d}{dt} (\phi(-t))|_{t=0} = \phi'(0) \\ \int_{-\infty}^{+\infty} -\delta'(x) \phi(x) dx = -[-\phi'(x)]|_{x=0} = \phi'(0) \end{cases} \Rightarrow \delta'(-x) = -\delta'(x)$$

$$(5) \begin{cases} \int_{-\infty}^{+\infty} x \delta'(x) \phi(x) dx = -\frac{d}{dx} [x \phi(x)]|_{x=0} = -\phi(0) \\ \int_{-\infty}^{+\infty} -\delta(x) \phi(x) dx = -\phi(0) \end{cases} \Rightarrow x \delta'(x) = -\delta(x)$$

3. 解: (1) $\begin{cases} u_t = a^2 u_{xx} & (0 < x < l, t > 0) \\ u(t, 0) = u(t, l) = 0 \\ u(0, x) = \delta(x - \frac{l}{2}) & (0 < x < l) \end{cases}$ 设 $u(t, x) = T(t) X(x)$, 代入方程有

$$\frac{T'(t)}{a^2 T(t)} = \frac{X''(x)}{X(x)} = -\lambda$$

$$X \text{ 方程: } \begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases} \Rightarrow \begin{cases} \lambda_n = (\frac{n\pi}{l})^2 \\ X_n(x) = \sin \frac{n\pi x}{l} \end{cases} \quad n \geq 1$$

将 λ_n 代入 $T'(t) + a^2 \lambda_n T(t) = 0$ 得 $T_n(t) = C_n e^{-a^2 (\frac{n\pi}{l})^2 t}$



扫描全能王 创建



中国科学技术大学

UNIVERSITY OF SCIENCE AND TECHNOLOGY OF CHINA

Hefei, Anhui. 230026 The People's Republic of China

有级数解: $u(t, x) = \sum_{n=1}^{+\infty} C_n e^{-\left(\frac{n\pi a}{L}\right)^2 t} \sin \frac{n\pi x}{L}$

代入 $u(0, x) = \delta(x - \frac{x_0}{2})$ 得 $C_n = \frac{2}{L} \int_0^L \delta(x - \frac{x_0}{2}) \sin \frac{n\pi x}{L} dx = \frac{2}{L} \sin \frac{n\pi x_0}{2L}$

故 $u(t, x) = \frac{2}{L} \sum_{n=1}^{+\infty} \sin \frac{n\pi x_0}{2L} \cdot \sin \frac{n\pi x}{L} e^{-\left(\frac{n\pi a}{L}\right)^2 t}$

5. 解: $\Delta_3 u(x) + k^2 u(x) = \delta(x)$ (作变换 $\hat{u}(\frac{x}{\rho}) = \int_{\mathbb{R}^3} u(x) e^{i\frac{x}{\rho} \cdot x} dx$)

$$\Rightarrow -|\frac{x}{\rho}|^2 \hat{u}(\frac{x}{\rho}) + k^2 \hat{u}(\frac{x}{\rho}) = 1 \Rightarrow \hat{u}(\frac{x}{\rho}) = \frac{1}{k^2 - |\frac{x}{\rho}|^2}$$

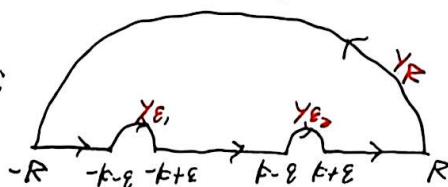
$$\begin{aligned} u(x) &= \frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} \hat{u}(\frac{x}{\rho}) e^{-i\frac{x}{\rho} \cdot x} d\frac{x}{\rho} \quad (\text{分别记 } r=|x|, \rho=|\frac{x}{\rho}|, \theta \text{ 为 } x, \frac{x}{\rho} \text{ 夹角;} \\ &= \frac{1}{8\pi^3} \int_0^{2\pi} d\varphi \int_0^{\infty} \frac{\rho^2}{k^2 - \rho^2} \int_0^{\pi} e^{-i\rho r \cos\theta} \sin\theta d\theta d\rho \quad (\text{球坐标, } d\frac{x}{\rho} = \rho^2 \sin\theta d\rho d\theta d\varphi) \\ &= \frac{1}{4\pi^2} \int_0^{+\infty} \frac{\rho^2}{k^2 - \rho^2} \cdot \left[\frac{e^{-i\rho r \cos\theta}}{-i\rho \cos\theta} \right] \Big|_{\theta=0}^{\theta=\pi} d\rho \end{aligned}$$

$$= \frac{1}{2\pi^2 r} \int_0^{+\infty} \frac{\rho \sin \rho r}{k^2 - \rho^2} d\rho$$

$$I := \int_0^{+\infty} \frac{\rho \sin \rho r}{k^2 - \rho^2} d\rho = \frac{1}{2} \int_{-\infty}^{+\infty} \frac{\rho \sin \rho r}{k^2 - \rho^2} d\rho = \frac{1}{2} \text{Im} \left(\int_{-\infty}^{+\infty} \frac{\rho e^{i\rho r}}{\rho^2 - k^2} d\rho \right) =: \frac{1}{2} \text{Im}(J)$$

$$\text{令 } f(z) = \frac{ze^{izr}}{z^2 - k^2}$$

取围道 Γ :



γ : x 轴上部分

$$\int_{\Gamma} f(z) dz = 0 \quad \text{BP}$$

$$(\int_{\gamma_{\epsilon_1}} + \int_{\gamma_{\epsilon_2}} + \int_{\gamma_R} + \int_{\gamma}) f(z) dz = 0$$

$$R \rightarrow +\infty \text{ 时, } \int_{\gamma_R} f(z) dz \rightarrow 0, \quad \epsilon \rightarrow 0 \text{ 时 } (\int_{\gamma_{\epsilon_1}} + \int_{\gamma_{\epsilon_2}}) f(z) dz \rightarrow \frac{i\pi}{2} e^{-ikr} + \frac{i\pi}{2} e^{ikr}$$

$$\Rightarrow \int_{-\infty}^{+\infty} f(z) dz = -\frac{i\pi}{2} e^{-ikr} - \frac{i\pi}{2} e^{ikr} = -i\pi \cos kr$$

$$\Rightarrow I = -\frac{\pi \cos kr}{2} \quad \text{代入有 } u(x) = -\frac{\cos kr}{4\pi r}$$





中国科学技术大学

UNIVERSITY OF SCIENCE AND TECHNOLOGY OF CHINA

Hefei, Anhui. 230026 The People's Republic of China

6.12) 解: $M(x, y, z)$ $M_0(\xi, \eta, \zeta)$ (原像 M_0 , 电荷: +1) $\rho_0 = \sqrt{\xi^2 + \eta^2 + \zeta^2}$

镜像: 球面对称 $M_1: (\frac{a^2\xi}{\rho_0^2}, \frac{a^2\eta}{\rho_0^2}, \frac{a^2\zeta}{\rho_0^2})$ 电荷: $-\frac{a}{\rho_0}$

平面 $z=0$ 对称: $M_2: (\xi, \eta, -\zeta)$ 电荷: -1

两次对称 $M_3: (\frac{a^2\xi}{\rho_0^2}, \frac{a^2\eta}{\rho_0^2}, \frac{-a^2\zeta}{\rho_0^2})$ 电荷: $+\frac{a}{\rho_0}$

$$G = \frac{1}{4\pi} \left[\frac{1}{r(M, M_0)} - \frac{a}{\rho_0} \frac{1}{r(M, M_1)} - \frac{1}{r(M, M_2)} + \frac{a}{\rho_0} \frac{1}{r(M, M_3)} \right]$$

